

1 Introduction

This document contains the following listings:

Listings

1	Another bit of Pascal	2
2	A C language listing	3
	any.sty.ltxml	4
	listing.tex	4

2 Inline Listings

Various delimiters: `a_word`, `a_word`, `a_word`, `a_word` and even `a_word` done.

Indirectly: `a_word`; and with messed up braces `foo { bar .`

Careful with spacing/math/macros: `foo(\langle X \rangle)`

2.1 Shorthands

Normal: `—@—` and x^y

Listing1: `$foo->baz(/^\s*/)"`;

Listing2: `$foo->baz(/^\s*/)"`;

Listing3: `xy`

Normal again: `—@—` and x^y

3 An untyped Listing

No options, language, etc

```
1 stuff1
2 stuff2
3 stuff3
```

4 Some C

```
1 #define EXAMPLE whichwhat
2 x = "foo";
3 break;
```

5 A Pascal Listing

A listing portion:

```
1 begin
2   { do nothing }
3 end;
```

A numbered listing:

```
1 for i:=maxint to 0 do
  ---- begin
3  ---- { do nothing }
  ---- end;

5 Write('case_insensitive');
7 Write('long_'_string');
Write('Pascal_keywords.');
```

A Titled listing:

A bit of Pascal

```
1 for i:=maxint to 0 do
2 begin
3   { do nothing }
4 end;
5 Write('case-insensitive');
```

A Captioned listing (known as Listing 1) :

Listing 1: Another bit of Pascal

```
100 for i:=maxint to 0 do
101 begin
102   { do nothing }
103 end;
```

6 An Environment

```
1 for i:=maxint to 0 do
2 begin
3   { do nothing }
4 end;
```

```
for i:=maxint to 0 do
begin
  { do nothing }
end;
```

1
2
3
4

for i:=maxint to 0 do	1
begin	2
{ do nothing }	3
end;	4

7 Framing and such

```

1 for i:=maxint to 0 do
2 begin
3   { do nothing }
4 end;
```

```

1 for i:=maxint to 0 do
2 begin
3   { do nothing }
4 end;
```

```

1 for i:=maxint to 0 do
2 begin
3   { do nothing }
4 end;
```

```

1 for i:=maxint to 0 do
2 begin
3   { do nothing }
4 end;
```

Listing 2: A C language listing

```

1 #define EXAMPLE whichwhat
2 x = "foo";
3 break;
```

8 Listing with Math

```

1 // calculate  $a_{ij}$ 
2 a[i][j] = a[j][j]/a[i][j];

1 // calculate  $a_{ij}$ 
2 a[i][j] = a[j][j]/a[i][j];
```

```

1  // calculate aij
2  aij = ajj/aij;
3  // calculate aij = sin x
4
5  a[i , j]=sin (x)
6  foo="a~word";
7  foo="a~x2 math";

1  // calculate <aij>
2  a_{ij}
3  = a_{jj}/a{ij};

1  // calculate $a_{ij}$
2  $a_{ij}
3  = a_{jj}/a{ij}$;
4  // calculate $a_{ij} =
5  \sin x$
6  a[i , j]=sin (x)
7  foo="a_~word";
8  foo="a_~\"string";
9  foo="a_~$x^2$_~math";

```

9 A Perl Listing

```

1  # -- CPERL --
2  package LaTeXML::Package::Pool;
3  use strict;
4  use LaTeXML::Package;
5
6  DefConstructor( '\container{}', "<ltx:special>#1</ltx:special>" );
7  DefConstructor( '\foo', "<ltx:not-defined/>" );
8
9  1;

```

10 A Recursive T_EX listing

```

1  \documentclass{article}
2  \usepackage{makeidx}
3  \makeindex
4  \usepackage{listings}
5  \usepackage[dvipsnames]{color}
6  \begin{document}
7  \lstset{numbers=left}
8  \section{Introduction}

```

```

9 This document contains the following listings:
10 \lstlistoflistings
11
12 \section{Inline Listings}
13 Various delimiters: \lstinline{a_word},
14 \lstinline!a_word!, \lstinline Aa_wordA,
15 \lstinline&a_word& and even \lstinline^a_word^ done.
16
17 \def\justcopy#1{#1}
18 Indirectly: \justcopy{\lstinline|a_word|};
19 and with messed up braces \lstinline{foo { bar }.% }
20
21 {
22 \lstset{
23     mathescape=true,
24 }
25 Careful with spacing/math/macros: \lstinline!foo($\langle X \rangle$)!
26 }
27 \subsection{Shorthands}
28 Normal: |@| and $x^y$
29 \lstMakeShortInline[language=perl,basicstyle=\ttfamily]|
30 \lstMakeShortInline[language=perl,basicstyle=\ttfamily]@
31 \lstMakeShortInline[language=perl,basicstyle=\ttfamily]^
32
33 Listing 1:
34 |$foo->baz(/^\s*/)";|
35
36 Listing 2:
37 @$foo->baz(/^\s*/)";@
38
39 Listing 3:
40 ^xy^
41 \lstDeleteShortInline|
42 \lstDeleteShortInline@
43 \lstDeleteShortInline^
44
45 Normal again: |@| and $x^y$
46
47 \section{An untyped Listing}
48 No options, language, etc
49 \begin{lstlisting}
50 stuff1
51 stuff2
52 stuff3
53 \end{lstlisting}
54

```

```

55 \section{Some C}
56
57 \begin{lstlisting}[language=C, identifierstyle=\slshape, directivestyle=\ttfamily]
58 #define EXAMPLE whichwhat
59 x = "foo";
60 break;
61 \end{lstlisting}
62
63 \section{A Pascal Listing}
64 A listing portion:
65 \begin{lstlisting}[language=Pascal, firstline=2, lastline=5, caption={}]
66 for i:=maxint to 0 do
67   begin
68     { do nothing }
69   end;
70
71   Write('case insensitive ');
72   Write('long ' ' string ');
73   WritE('Pascal keywords. ');
74 \end{lstlisting}
75
76 A numbered listing:
77 \begin{lstlisting}[language=Pascal, numbers=left, numberstyle=\tiny, stepnumber=2]
78 for i:=maxint to 0 do
79   begin
80     { do nothing }
81   end;
82
83   Write('case insensitive ');
84   Write('long ' ' string ');
85   WritE('Pascal keywords. ');
86 \end{lstlisting}
87
88 A Titled listing:
89 \begin{lstlisting}[language=Pascal, title={A bit of Pascal}]
90 for i:=maxint to 0 do
91   begin
92     { do nothing }
93   end;
94   Write('case insensitive ');
95 \end{lstlisting}
96
97
98 A Captioned listing (known as Listing \ref{pascallisting}) :
99 \begin{lstlisting}[language=Pascal, caption=Another bit of Pascal, label=pascallisting]
100 for i:=maxint to 0 do

```

```

101 begin
102   { do nothing }
103 end;
104 \end{lstlisting}
105
106 \section{An Environment}
107 \begin{lstlisting}[language=Pascal]
108 for i:=maxint to 0 do
109   begin
110     { do nothing }
111   end;
112 \end{lstlisting}
113
114 \lstnewenvironment{colored}[1]{\lstset{language=Pascal,numbers=right,numberstyle=
115 \begin{colored}{red}
116 for i:=maxint to 0 do
117   begin
118     { do nothing }
119   end;
120 \end{colored}
121
122 \begin{colored}{blue}
123 for i:=maxint to 0 do
124   begin
125     { do nothing }
126   end;
127 \end{colored}
128
129 \section{Framing and such}
130 \lstset{backgroundcolor=\color[named]{CarnationPink}}
131 \begin{lstlisting}[language=Pascal,frame=single,rulecolor=\color{red}]
132 for i:=maxint to 0 do
133   begin
134     { do nothing }
135   end;
136 \end{lstlisting}
137
138 \begin{lstlisting}[language=Pascal,frameround=tttt,backgroundcolor=\color{yellow}
139 for i:=maxint to 0 do
140   begin
141     { do nothing }
142   end;
143 \end{lstlisting}
144 \lstset{backgroundcolor=}
145 \begin{lstlisting}[language=Pascal,frame=single]
146 for i:=maxint to 0 do

```

```

147 begin
148   { do nothing }
149 end;
150 \end{lstlisting}
151
152 \begin{lstlisting}[language=Pascal,frame=lines]
153 for i:=maxint to 0 do
154   begin
155     { do nothing }
156   end;
157 \end{lstlisting}
158
159 \begin{lstlisting}[language=C,identifierstyle=\slshape,directivestyle=\ttfamily,
160 caption=A C language listing, frame=lines,backgroundcolor={\color[cmym]{0,0,0,0.
161 #define EXAMPLE whichwhat
162 x = "foo";
163 break;
164 \end{lstlisting}
165
166 \section{Listing with Math}
167 \begin{lstlisting}[language=c, texcl, commentstyle=\color{green}]
168 // \upshape calculate $a_{ij}$
169 a[i][j] = a[j][j]/a[i][j];
170 \end{lstlisting}
171
172 \begin{lstlisting}[texcl, language=c]
173 // \upshape calculate $a_{ij}$
174 a[i][j] = a[j][j]/a[i][j];
175 \end{lstlisting}
176
177 \begin{lstlisting}[language=c, mathescape, numbers=left, commentstyle=\color{green}]
178 // calculate $a_{ij}$
179 $a_{ij}
180 = a_{jj}/a_{ij}$;
181 // calculate $a_{ij} =
182 \sin x$
183 a[i,j]=sin(x)
184 foo="a word";
185 foo="a $x^2$ math";
186 \end{lstlisting}
187
188 \begin{lstlisting}[language=c, escapechar=\%, escapebegin=\textless, escapeend=\tex
189 // calculate %$a_{ij}$%
190 a_{ij}
191 = a_{jj}/a_{ij};
192 \end{lstlisting}

```



```

193
194 \begin{lstlisting}[language=c,numbers=left,stringstyle=\ttfamily]
195 // calculate $a_{ij}$
196 $a_{ij}
197 = a_{jj}/a_{ij}$;
198 // calculate $a_{ij} =
199 \sin x$
200 a[i,j]=sin(x)
201 foo="a word";
202 foo="a \"string\"";
203 foo="a $x^2$ math";
204 \end{lstlisting}
205
206 \section{A Perl Listing}
207 \lstinputlisting[language=perl]{any.sty.ltxml}
208
209 \section{A Recursive \TeX\ listing}
210 \lstinputlisting[language={[LaTeX]TeX}]{listing.tex}
211
212 A shorter listing, with colored cs that include the slash
213 \begin{lstlisting}[language={[LaTeX]TeX},texcsstyle={\color{blue}\bfseries}]
214 \iftrue something \fi
215 \end{lstlisting}
216
217 \section{Testing Tag}
218 % AHA, tagstyle only is in effect with XML (?)
219 \begin{lstlisting}[language=XML,tagstyle=\bf]
220 <element attr='value'>content</element>
221 \end{lstlisting}
222 \begin{lstlisting}[language=XML,tagstyle=\bf,usekeywordsintag=false]
223 <element attr='value'>content</element>
224 \end{lstlisting}
225 \begin{lstlisting}[language=XML,tagstyle=\bf,markfirstintag]
226 <element attr='value'>content</element>
227 \end{lstlisting}
228
229 \section{Literate Programming}
230 \begin{lstlisting}[language=C,escapechar=@,literate={:=}{\${\gets$}}1 {<=}{\${\le$}}]
231 var i:integer;
232 if (i<=0) i := 1;@\label{lit:a}@
233 if (i>=0) i := 0;
234 if (i<>0) i := 0;@\label{lit:b}@
235 /* However not := here */
236 \end{lstlisting}
237 where we draw your attention to lines \ref{lit:a} and \ref{lit:b}.
238

```

```

239 \section{Screwiness}
240 \lstdefinelanguage{bingo}{\morekeywords={foo , bar } , \morekeywords=[2]{bing , bar }}
241 %,
242 % AHA, words can only be in one class (1st one declared?)
243 % BUT, index is separate, and classname is without the "style" !!
244 \begin{lstlisting}[language=bingo , keywordstyle=\bfseries\underbar , keywordstyle=
245 foo bar baz bing booboo
246 \end{lstlisting}
247 {\bfseries\itshape bfit}
248 {\itshape\bfseries itbf}
249 \printindex
250 \end{document}

```

A shorter listing, with colored cs that include the slash

```

1 \iftrue something \fi

```

11 Testing Tag

```

1 <element attr='value'>content</element>
1 <element attr='value'>content</element>
1 <element attr='value'>content</element>

```

12 Literate Programming

```

1 var i:integer;
2 if (i≤0) i ← 1;
3 if (i≥0) i ← 0;
4 if (i≠0) i ← 0;
5 /* However not := here */

```

where we draw your attention to lines 2 and 4.

13 Screwiness

```

1 foo bar baz bing booboo
bfit itbf

```